

ECEN 607: Advanced Analog Circuit Tech

Lab6: a) Multi-Stage Op-Amp with Nested Miller Compensation b) Using Cascoded Miller Compensation and Ahuja Compensation

Pre-lab or Lab Report

Name: FirstName LastName
UIN: ***Last Three Digits**

Objectives

- To design multi-stage amplifiers using nested miller compensation
- To study cascoded Miller compensation and Ahuja compensation

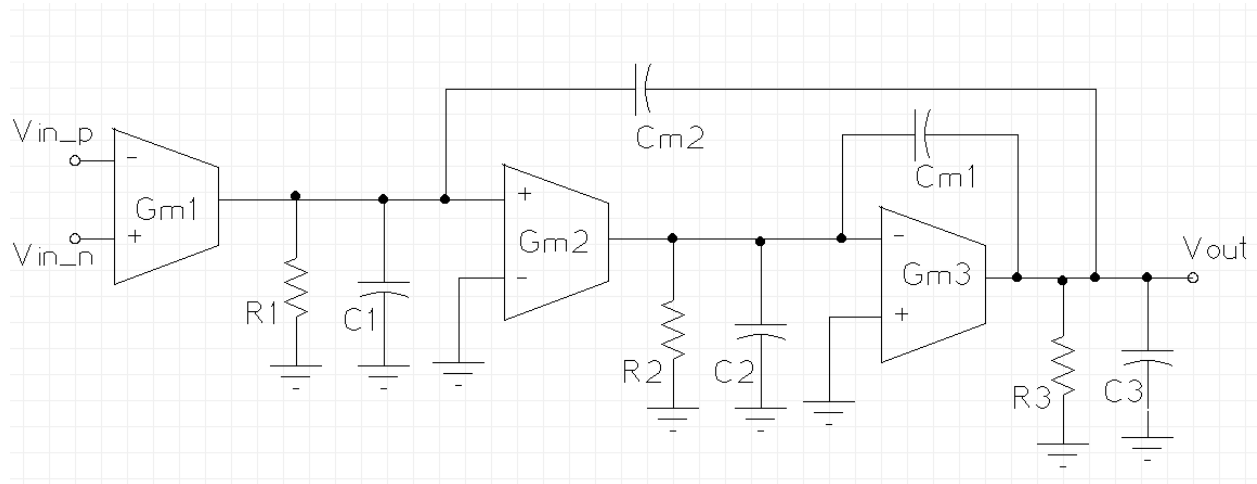


Figure 7-1: Three stage op amp with nested miller compensation

The above figure shows a 3-stage op amp with nested miller compensation at the transconductor level. R and C at each transconductor output represent the equivalent impedances at that node. Cm1 and Cm2 are the miller capacitances. The dominant pole is set at the output of the first stage while the second pole is determined by Cm1 and the gain of Gm3 and its loading.

Instead of connecting directly the miller capacitor, the Ahuja's compensation approach uses a low-impedance node to feed back the current generated by the miller capacitor; the capacitor is still connected to the output to generate a current that is proportional to the output voltage only.

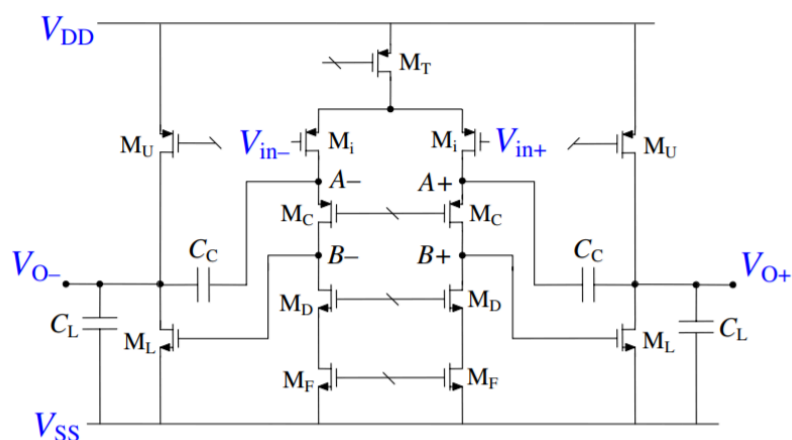


Figure 7-2: An example showing cascoded miller compensation employing Ahuja's approach

The conventional miller capacitor generates a *current injected to the output node* given by $i_c = sC_m(v_{in} - v_{out})$; Ahuja noticed in his paper that this current component is detrimental since it adds a

capacitance at the output node due to the component $sC_m(-v_{out})$, and generates the RHP zero due to the non-inverting part $sC_m(v_{in})$. To eliminate this term, the compensating capacitor is connected to a low-impedance node instead of directly to v_{in} . At the input side v_{in} , the current component given by $sC_m(v_{in})$ adds capacitance at this node while the component. The current component (injected into node v_{in}) $sC_m(v_{out})$ corresponds to the desired miller compensation. These goals are achieved if the capacitor is connected between amplifier's output v_{out} and a low-impedance node such as a cascode node. The miller current reflected in both terminals can then be approximated as $sC_m(-v_{out})$ instead of $sC_m(v_{in} - v_{out})$. Thus the detrimental RHP zero is eliminated. The realization is illustrated in Fig. 7.2; the miller capacitor C_c is connected between output node and source of M_c (low gain node A) instead of connecting it between V_o and node B. Compensating current is mainly generated by $C_c v_{out}$. Notice that

the current of C_c is channelized to node B (output of the first stage) through the source of M_c ; compensating current reaches the output of the first amplifier. Notice that same effect, and probably more effective, is obtained if the compensating capacitor is connected through the cascode node (source terminal) of the N-type transistors M_b .

The effect is better illustrated in Fig. 7.3 where C_f is coupled to amplifier's input through a P-MOS transistor. To make practical this approach, use cascode nodes usually embedded in high gain amplifiers.

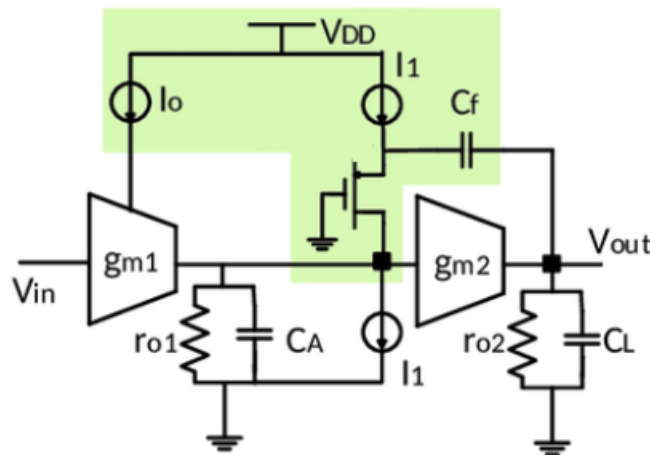


Figure 7-3: Ahuja compensation for a two-stage op amp (at transconductor level) with a dedicated transistor

Prelab

1. In this prelab, use 1.8V NFETs and PFETs.
 - Do not forget to show a detailed design procedure for each circuit.
 - Include a summary table with all transistor dimensions and biasing conditions.
1. Design the single ended nested miller op amp (Fig7-1) to meet the specifications shown in Table 7-1. The stage 1 transconductor should be cascoded (you may choose to cascode the load/stage2/stage3 or not). Assume only current source I_{bias} (max 5 uA) is available, i.e. you need to design biasing circuits. Also, you are not allowed to move the miller zeroes to the LHP.

Table 7-1: Design specifications

Supply Voltage range (USB spec)	$1.8 \pm 5\%$
A_{vo}	100dB
C load	10 pF
Unity Gain Frequency (UGF)	>5 MHz
Slew Rate	> 5 V/us
PM	> 50°
Output Swing	1V
Quiescent Power	2 mW
Temperature range	-40 C to 125 C

Draw the schematic of the op amp neatly with W&L values of transistors shown. Also, show the current in each branch, bias voltages if any etc.

2. Fig.7-2 shows a fully differential two-stage op amp (stage1 is telescopic, stage2 is common source). Usually, the miller caps (C_c) are connected to nodes B+ and B-. In this figure, it is connected to the cascoding node (A and A+) to mitigate the RHP zero effect and is commonly known as cascoded miller compensation.
 - a. How does the compensation work here? Show relevant equations.
 - b. What are the advantages and disadvantages of this method?
 - c. Explain the scenario during slew (positive and negative)

Lab Report

Figure7-1: Nested Miller Opamp

1. Simulate the circuit from the prelab (the nested miller op amp) and adjust the transistor sizes accordingly until you meet all specifications. Provide the necessary plots such that you show clearly that you meet all the specs. Measure the input offset voltage.
2. Keep the op amp in unity buffer config and report the HD2, HD3, THD linearity metrics of the system. Use a sinusoidal input of 50 kHz frequency, 100 mVpeak amplitude for this. Use PSS or FFT based approach for this.
3. Show the step response of the op-amp. Keep the op-amp in unity buffer config itself. Apply a step of +/- 100 mV and measure the 7τ settling time. Compare it with expected settling time. If you see a difference, explain why.
4. Provide comparison tables required specs vs. simulated specs.
5. Comment on your results and provide conclusions.

Figure7-2: Ahuja Compensation:

6. Redesign your amplifier using Fig.7-2, and repeat items 1-5.
7. Compare your results with the conventional Miller case. What are your observations?
8. Report the pulse response; 100mV input pulse, falling and rising 1% settling time.
9. Show +PSRR, -PSRR plots and measure the input offset voltage & input referred noise (10Hz - GBW)
- 10.** Provide comparison tables for op-amps – report only DCgain, phase margin, GBW, PSRRs, Input offset voltage, Input referred noise. Comment on your results and provide conclusions.